

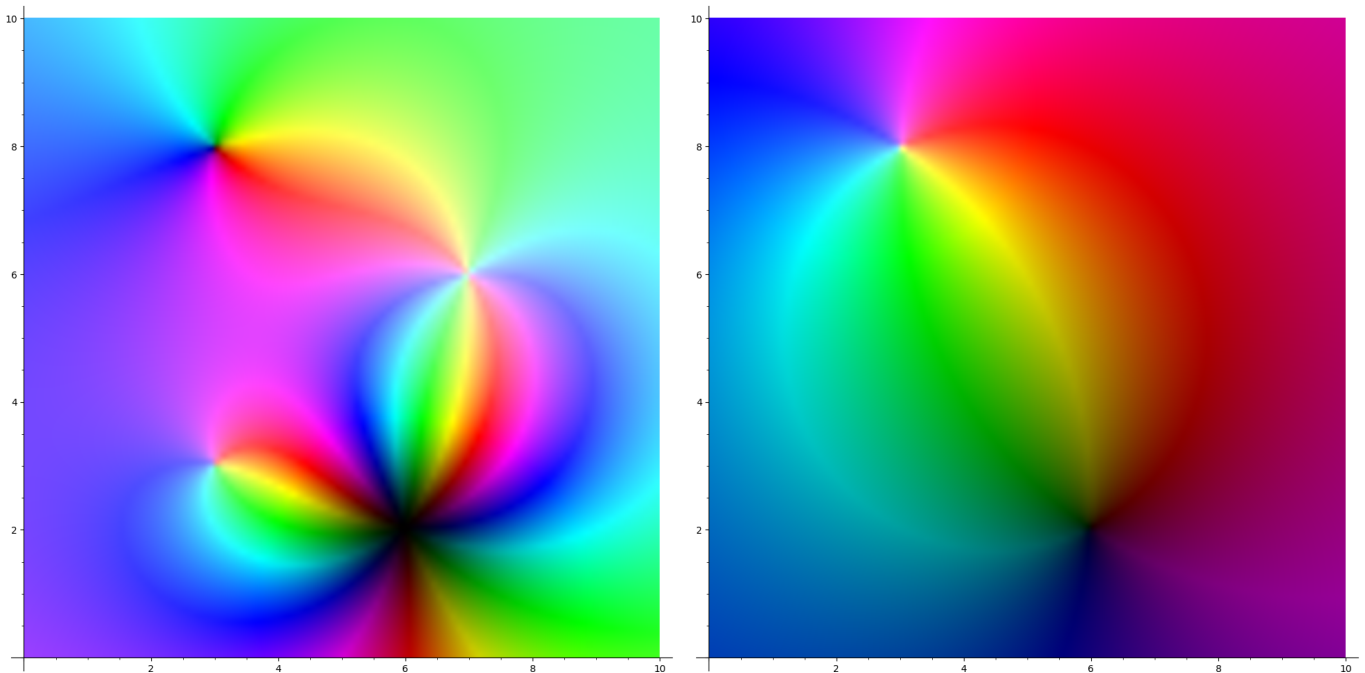
Complex Variable - Midterm Exam

考試時間 Test time 10:30:00 am to 12:00:00 pm; 1.5 hours. 以教室投影機時鐘為準 Using projected clock as official time. 用答案紙及空白紙張作答 Answer on the answer sheet and then on blank paper. 每題五分 Five points per problem. 可以帶 991 計算機 You can use fx-991 or anything weaker. 舉手問問題 Raise your hand to ask questions.

[1] Find a subset S of the complex plane $\mathbb{C}$ such that S, S', S'', S''', S'''' are all different. Here, S' is the derived set (the collection of all limit points) of S . You get one point per unique derived set. That is, your score is $|\{S, S', S'', S''', S''''\}|$.

[2] $T \subset \mathbb{C}$ is an infinite subset such that $|t| < 10$ for any $t \in T$. Show that the derived set of T is nonempty. For this question, you can use the infinity version of the pigeonhole principle. Usually, the pigeonhole principle states that if we try to put $p + 1$ pigeons into p holes, then at least one hole contains two pigeons or more. But if we have infinitely many pigeons and p holes, then at least one hole contains infinitely many pigeons.

[3] Consider the left figure. Perform the following actions on the answer sheet. Circle all the roots. If it is a double root, circle twice, and so on. For a pole, draw triangles instead of circles. Mark positive real images (pure red) with solid lines. Mark negative real images (pure cyan) with dotted lines.



[4] Consider the right figure. It is the map

$$z \mapsto \frac{(a + bi) + (c + di)z}{(e + fi) + (g + hi)z}$$

for some integers (a, b, c, d, e, f, g, h) whose greatest common divisor is 1, and e and f are positive. Find those integers. One point per correct integer after the first three correct integers.

[5] What conformal map maps the strip $1 < x + 2y < 2$ to the fan $1 < \arg z < 2$?

[6] A bijection is a map that is one-to-one (injective) and onto (surjective). The map $\beta(z) = (z - i)/(z + i)$ maps the upper half plane $y > 0$ to the unit disk $r < 1$ bijectively. (This is because, as z is closer to i than to $-i$, we deduce that $|z - i|$ is smaller than $|z + i|$ and hence $|z - i|/|z + i| < 1$.) Find the bijective map that maps the unit disk to itself, 1 to 1, and 0 to $1/2$. There is only one such function. Just find the function. You don't have to prove uniqueness.

[7] Show that, for any complex number $z \neq 0$, the sum $\sum_{m=1}^{\infty} m!z^m$ does not converge.

[8] The Riemann zeta function is defined as

$$\zeta(z) = \sum_{n=1}^{\infty} \frac{1}{n^z}$$

for all complex numbers z with real part > 1 . We know that $\zeta(5+5i) \approx 0.?????9242554574 + 0.0119825174357425i$. Find the missing digits. One point per digit.

[9] Define $\kappa(z) = \cos(\sin(\cos(\sin(\cos(\sin(z)))))$). Its second derivative at 0 is $-0.?????0487021611$. Find the missing digits. One point per digit.

[10] Evaluate the imaginary part of the definite integral $\int_{\gamma} \exp(z^2) dz$ where the curve γ is a half circle starting from i , passing through $1 + 2i$, and ending at $3i$. The answer is $0.?????3215447094$. Find the missing digits. One point per digit.

[11] Let the curve $\delta(\tau)$ be $(r, \theta) = (2 \sin(2\tau), \tau)$ for τ from 0 to 2π . Evaluate

$$\oint_{\delta} \frac{\cos(z) dz}{z^8 - 1}.$$

[12] The integral below is about $0.?????5218126363$. Find the missing digits. One point per digit.

$$2 \int_0^{\infty} \frac{\cos(x) dx}{x^2 + \sqrt{2}}$$

Hint: It happens that $\cos(x)$ is the real part of e^{ix} . So the problem is equivalent to the integral with e^{ix} in the numerator, and discarding the imaginary part after finishing. Hint 2: Now integrating over the real line is the limiting case of integrating over a long interval $[-M, M]$ and then the half circle passing M , Mi , and Mi^2 . How come we can add the half circle out of the thin air? This is essentially because the integrand is $O(1/M^2)$. For this question, just give me the evaluation of the integral.

[13] For this question, show me why the contribution of the half circle tends to zero as M tends to infinity.

[14] A function $f : \mathbb{C} \setminus \{0\} \rightarrow \mathbb{C}$ is holomorphic on the punctured plane $\mathbb{C} \setminus \{0\}$. It is said to have a removable singularity at 0 if $\lim_{z \rightarrow 0} f(z)$ exists. Now we try to “patch” the singularity at 0 by defining

$$g(z) = \begin{cases} f(z), & z \neq 0 \\ \lim_{z \rightarrow 0} f(z), & z = 0 \end{cases}$$

Show that g is holomorphic on $\mathbb{C}$. Hint: The problem is that g is only continuous at 0 and may not be differentiable at 0. After multiplying g by something magic, we can make it such that $(\text{magic} \cdot g)'(0)$ exists and we know exactly its value. And then we can divide the magical term and get back to g . Do not cite big theorems. We are trying to reproduce the proof.

[15] $h : \mathbb{D} \rightarrow \mathbb{D}$ is holomorphic on the unit disk $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ such that $h(0) = 0$ and $|h(z)| \leq 1$. Show that $|h(z)| \leq |z|$.

[16] A function $u : \mathbb{R}^2 \rightarrow \mathbb{R}$ is harmonic if its Laplacian $(\partial^2 u / \partial x^2) + (\partial^2 u / \partial y^2)$ is zero everywhere. For such a function, there exists another function $v : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that $u + iv$, as a function in (x, y) , is holomorphic in $z = x + yi$. Try to express v in terms of some beautiful integral of something about u (3 points). Give a brief (about 50–100 words) explanation why your $u + iv$ is holomorphic.

[17] For a sequence $a_0, a_1, a_2, \dots$, its generating function is defined as $G(z) = a_0 + a_1 z + a_2 z^2 + \dots$. Suppose that $a_0 = 0$ and $a_1 = 2$, and that $4a_{j-1} - 4a_j + a_{j+1} = 0$ for all j that apply. Find the generating function of this sequence (2 points). Determine its radius of convergence (1 point). Find the generating function of the filtered sequence $a_0, 0, 0, a_3, 0, 0, a_6, 0, 0, a_9, \dots$ (2 points).

[18] For these three problems, one point per blank. Answer on the answer sheet or blank paper. By the _____ theorem of _____, the polynomial $z^3 - 5iz + 20$ has _____ roots. Note that when $|z| > r = ______ (an integer)$, $|z^3|$ is greater than $|5iz| + |20|$, which is greater than or equal to $|-5iz + 20|$ by the _____ inequality. So there is no root outside the disk of radius r .

[19] We can also see that there is one root in the _____ quadrant, one in the _____ quadrant, and one in the _____ quadrant. Now using Newton's iteration $z_{n+1} = z_n - \frac{f(z_n)}{f'(z_n)}$, we can quickly approximate the roots.

[20] The roots are, approximately, _____4673326721+_____27885641441i, _____34570814979+_____3009971984i, and _____1216245223 + _____5798536128i.

[21] One point: Write this down on the answer sheet or blank paper: *I do not cheat on the exam, I take photos of my exam paper after the exam time ends, and I upload those photos to NTU COOL for future reference.*

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Name: _____ Student ID: _____

